

Yves De Boeck

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TECHNICAL SKILLS

MLOps & Data Infrastructure: MLflow, Data & ML Lifecycle, Azure ML/Data Services, Databricks, Scalable Data Pipelines, Pipeline Observability, Dataset Quality Control

Robotics & Edge Systems: Embedded C++ (Yocto OS), MQTT, IoT Hub, Sensor Telemetry, Real-time Streaming, Edge-to-Cloud Communication, Arduino, 3D Printing

ML Models & AI: NeuralProphet (Time-Series Forecasting), PyTorch/TensorFlow (ML Concepts), Vector Databases, RAG Architecture, Azure Agent Framework, Azure AI Foundry

Software Engineering & CI/CD: Python (FastAPI, Mypy, Poetry, UV), SQL, Docker, Kubernetes, Terraform (IaC), Bicep, Azure DevOps, Jenkins, GitLab CI, Git, TDD (pytest, gtest)

Engineering Foundations: Distributed Systems, ETL/ELT (Batch & Streaming), SOLID Principles, Object-Oriented Programming, Design Patterns, Agile/Scrum

EXPERIENCE

Senior Data and Mlops Engineer

Jan. 2024 – Present

Atlas Copco

Antwerp, Belgium

- Architected a **production-scale lakehouse data platform** using **Databricks (Unity Catalog)**, **Delta Lake** and **dbt**, enabling automated ETL and data integrity at scale for a diverse set of data sources ranging from manufacturing data, over IOT telemetry to CRM systems.
- Designed and implemented data and **MLOps** infra for a time-series forecasting model (NeuralProphet), utilizing automated workflows in Azure Databricks and MLflow to maintain model accuracy, dataset quality and manage the model life cycle in general. Deployment and live inference of this model resulted in 15% decrease in operational energy costs to our customers.
- Developed a **C++ communication component** as part of embedded firmware (Yocto OS), managing bidirectional data flow via **MQTT** between remote devices and our cloud (**Azure IoT Hub**)
- Owned and **optimized end-to-end data pipelines** for high volume batch and streaming workloads, at times slicing existing pipeline run times by up to 80% using **spark** optimization.
- Architected a **real-time streaming cloud infrastructure** using WebSockets to visualize live telemetry, enabling technicians to perform remote diagnostics and firmware updates as part of a large **Python/React** based fleet management platform.
- Built Teams-integrated **Agentic LLM application** that function as a 'digital colleague' using **Azure Agent Framework** and **Azure AI Foundry**, including automated security checks, AI flow evaluation and observability.
- All items above were always implemented with software engineering best practices applied, including extensive, automated test suites, IaC (**Terraform and Bicep**) and CI/CD (**Azure DevOps/Jenkins**).

Data / Software Engineer

Sept. 2021 – Dec. 2023

TrinetX

Ghent, Belgium

- Designed high-availability microservices and data-intensive applications in a Linux-based environment, leveraging **Docker and Kubernetes** for scalable deployment.
- Built a new internal service from scratch to monitor complex **ETL health**, identifying and resolving processing bottlenecks for multi-sensor healthcare data sources.
- Automated deployment workflows using Terraform and Gitlab pipelines, adhering to strict **TDD and unit testing** standards to ensure mission-critical software quality.

Data / Software Engineer (Billing Processes)

Sept. 2020 – Sept. 2021

Essent

Belgium

- Tracked down and resolved complex issues and anomalies in a multi-language custom SAP system leveraging and optimizing advanced SQL queries.

CERTIFICATIONS

Databricks: Data Engineering Professional, Data Engineering Associate, Machine Learning Associate (In Progress)

Microsoft Azure: Data Engineer Associate (DP-203), AI Engineer Associate (AI-102), Azure Fundamentals (AZ-900), Data Fundamentals (DP-900), AI Fundamentals (AI-900)

Cloud & Agile Architecture: AWS Certified Solutions Architect, Professional Scrum Master (Scrum.org - PSM I)

Programming: Oracle Certified Java SE 8 Professional, Python Institute Certified Associate in Python

EDUCATION

University of Antwerp

Antwerp, Belgium

Master of Science in Engineering: Electronics and ICT (Magna Cum Laude)

2016 – 2020